

Centralized Crypto Exchange

Feature Requirements Document

1. Core Trading Features:

- Spot Trading
- Derivatives Trading
- Perpetual Futures
- Options Trading
- Standard Futures Trading
- Crypto-to-Crypto Convert (Instant Swap)
- Copy Trading - OTC Trading Desk
- Multichain USDT Support
- Native Platform Token Support (Utility Token)

2. Liquidity & Order Book Requirements

- Aggregated order book integration from: - Binance - Bybit - Two additional exchanges (to be finalized)
- Internal exchange liquidity pool
- Fully in-house order matching engine with configurable matching logic (price-time priority)

3. Market Infrastructure

- P2P Marketplace (Buy/Sell Crypto with FIAT)
- Staking & Earn Products
- IEO/IDO Launchpad
- VIP Account Tiers with fee discounts and privileges

4. Compliance & Security

- Automated third party KYC/AML integration (global vendors)
- Travel Rule compliance provider integration
- Geo fencing & jurisdiction control
- 3rd party custodial wallet integration

- Secure hot/cold wallet system
- Anti-fraud systems & transaction monitoring

5. Deposits & Withdrawals

- Crypto deposits & withdrawals (multi-chain)
- Fiat on ramp and off ramp integrations (via partners)
- Withdrawal whitelist options
- Automated fee calculation based on blockchain

6. Platform Applications

- Web App (Responsive)
- Android App (Native)
- IOS App (Native)
- Multi language support
- Push notifications, in-app alerts (price alert, updates, marketing banners)

7. Admin & Operations

- Super Admin Panel
- Sub Admin / Multi role Access Control (RBAC)
- Trade monitoring dashboard
- Dispute resolution dashboard (for P2P)
- Analytics, reporting & logs
- Content management system (for banners, notifications, updates)

8. User Features

- Referral & Affiliate Program
- Fee discount logic (VIP, token-based, volume-based)
- Watchlist, price alerts
- Notification center
- KYC status tracking
- Order history, trade history
- Account activity logs

9. Extra Modules

- NFT Marketplace (optional but requested)
- Earn products (Staking / Locked Staking / Flexible Earn)

- API trading support (REST + WebSocket)
- Market making bot connectivity

10. Performance Requirements

- High frequency, low latency order matching engine
- Scalable microservices based backend architecture
- Modular API first system
- High availability infrastructure (99.9% uptime target)

11. Security Requirements

- 2FA (Google Authenticator / SMS)
- Biometric authentication (mobile)
- Anti phishing code
- Device management
- IP whitelisting - Threshold & risk-based withdrawal approvals

12. Documentation Requirements

- API documentation (developer-friendly)
- Admin panel user documentation
- Compliance manuals & workflow docs